

# Final Project: Build an Isometric Room in Blender

## 1. Project Overview

For this final project, you will create a detailed **isometric 3D room** scene in Blender. An isometric room is viewed from a fixed corner angle with no perspective distortion, giving it a clean, illustrative, game-like feel. You will model all the furniture and decorations yourself using the tools and modifiers you have learned this term.



**Theme:** “*My Cozy Room*” – a small bedroom, study, or creative workspace.

## 2. Learning Objectives

By completing this project, you will demonstrate that you can:

- Plan and block out a complete 3D scene from scratch.
- Use **extrude, inset, bevel, subdivision surface, and mesh effects** (like Solidify or Wireframe) appropriately.
- Maintain clean topology and good object proportions.
- Set up an **isometric camera** and arrange objects to achieve a strong composition.
- Apply **simple materials and lighting** to enhance the final render.

## 3. Technical Requirements (What You Must Use)

Your scene must contain clear examples of the following tools/modifiers. You will label these in your submission.

| Requirement                              | Where you might use it                                                                                                                                                                                                                                                                |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extrude</b>                           | Pulling out desk legs, books from a flat face, lamp arms, window frames                                                                                                                                                                                                               |
| <b>Inset</b>                             | Creating the screen area of a computer monitor, picture frames, cupboard panels                                                                                                                                                                                                       |
| <b>Bevel</b>                             | Softening the edges of a tabletop, bed, books, or any hard-surface object                                                                                                                                                                                                             |
| <b>Subdivision Surface</b>               | Smoothing a cushion, a beanbag, a soft toy, or rounded furniture                                                                                                                                                                                                                      |
| <b>Mesh Effect (choose at least one)</b> | <ul style="list-style-type: none"> <li>• <b>Solidify</b> to give thickness to a rug, curtains, or a poster on the wall</li> <li>• <b>Wireframe</b> to create a decorative basket or geometric lamp</li> <li>• <b>Array</b> to make a row of books or a tiled floor (bonus)</li> </ul> |

Additionally, you must:

- Set up an **orthographic isometric camera** (instructions below).
- Include **at least one light source** (e.g., a Sun lamp, or an Area light to simulate a window).

## 4. Isometric Camera Setup (Step-by-Step)

Follow these steps exactly to achieve the classic isometric look:

1. Delete the default camera and add a new one: **Add** → **Camera**.
2. With the camera selected, go to the **Object Data Properties** (green camera icon).
3. Change **Type** from *Perspective* to **Orthographic**.
4. Set the **Orthographic Scale** to a value that frames your room well (start with 8 to 12, then adjust).
5. In the 3D Viewport, press **N** to open the side panel. With the camera still selected, set its **Rotation** to:
  - X: 60°
  - Y: 0°
  - Z: 45°

*(This rotates the camera to the classic isometric angle.)*
6. Move the camera (Location) to a position like (10, -10, 10) so it looks down into the room corner, then adjust until you like the view.
7. Lock the camera to this view by pressing **Numpad 0** (or **View** → **Cameras** → **Active Camera**).

**Important:** Do not change the camera's rotation after this setup – all your scene composition must work from this exact angle.

## 5. Suggested Workflow

| Phase                                                      | What to do                                                                                                                                                                                                         |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Reference &amp; Sketch</b> (before you open Blender) | Collect 3–5 real isometric room images you like. Draw a simple top-down sketch showing where the walls, floor, and major furniture will go.                                                                        |
| <b>2. Blockout</b>                                         | In Blender, build the room shell (floor and two walls meeting at a corner) using simple cubes. Scale them roughly to size. Add placeholder cubes for bed, desk, shelf, etc.                                        |
| <b>3. Modeling</b>                                         | One by one, replace each placeholder with your detailed model. Use the required tools from the table above. <b>Don't forget to add bevels</b> – they make objects look real.                                       |
| <b>4. Materials &amp; Colors</b>                           | Open the Shading workspace. Assign a new material to each main object. Change only the <b>Base Color</b> and <b>Roughness</b> for now. Keep the palette limited to 3–5 main colors to make the room feel cohesive. |
| <b>5. Lighting</b>                                         | Add a <b>Sun lamp</b> to create soft shadows across the room (rotate it to simulate light coming from a window). Optionally, add a dim Area light inside to brighten dark corners.                                 |
| <b>6. Final Composition</b>                                | Review the camera view. Adjust object positions so nothing clips awkwardly. Check that the most important items are clearly visible.                                                                               |

## 7. Render Settings

In the Render Properties, choose **Cycles** or **Eevee**. Set output resolution to 1920 x 1920 pixels (square works well for isometric). Turn on **Ambient Occlusion** for extra depth.

## 6. Required Objects (Minimum)

Your room must include **all of the following**:

1. **Floor and two walls** – the corner of a room.
2. **A bed or a desk/table** – the main piece of furniture.
3. **One piece of wall decor** – a poster, painting, window, or shelf.
4. **A rug or mat** (use Solidify modifier to give it thickness).
5. **A small technology item** – a computer monitor, laptop, or game console.
6. **At least two small props** – books, a mug, a plant pot, a pen, a clock, etc.

You are strongly encouraged to add extra personal details (posters, toys, musical instruments, etc.) that show your creativity – these will count toward the “Creativity & Effort” grade.

## 7. Deliverables

Submit **THREE** file containing:

- Your Blender project file (yourusername\_isometric\_room.blend).
- One final render: yourname\_isometric\_room.png (1920 x 1920 pixels).
- A short **PDF or text document** that lists:
  - Where you used each required technique (e.g., “*I used Inset on the monitor screen and bevel on the desk edges.*”)
  - Your self-reflection: one thing you are proud of, and one thing you would improve with more time.

## 8. Grading Rubric (100 points)

| Criteria                                       | Weight | Expectation                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Integrity</b>                               | 30%    | Submit your own work. No plagiarism is accepted.                                                                                                                                                                                                                                                                           |
| <b>Modeling &amp; Tool Use</b>                 | 25%    | Extrude, Inset, Bevel, Subdivision Surface, and at least one mesh effect (such as Solidify or Wireframe) are used clearly and appropriately. Models are well-built with clean topology, good proportions, and thoughtful detail.                                                                                           |
| <b>Scene Composition &amp; Isometric Setup</b> | 10%    | The camera is correctly set to an orthographic isometric view (accurate rotation angle, appropriate orthographic scale). Objects are intentionally arranged to create a visually balanced composition with a clear focal point. The frame feels full but not cluttered, making effective use of the limited viewing angle. |
| <b>Materials &amp; Lighting</b>                | 10%    | More than three base materials are used with deliberate color choices. The overall palette is harmonious and cohesive. Lighting adds depth and atmosphere; shadows are soft and natural, and details in darker areas remain readable.                                                                                      |
| <b>Completion &amp; Requirements</b>           | 10%    | All required objects (floor and two walls, main piece of furniture, wall decor, rug, tech item, and at least two small props) are fully present. Extra personal details are added beyond the minimum, showing strong effort. All submission files are complete and properly organized.                                     |
| <b>Creativity &amp; Effort</b>                 | 15%    | The room shows a distinct personal style and a unified theme. Unexpected or playful details are included, making the scene feel alive and full of storytelling. A strong sense of creative enthusiasm is evident throughout the work.                                                                                      |

**Total: /100**

## 9. Project Timeline (Suggested)

- **Week 1:** Reference gathering, sketch, blockout done.
- **Week 2:** All main furniture modeled.
- **Week 3:** Props, materials, lighting, and first test render.
- **Due date:** Jun 7th, 11:59 PM.

## 10. Tips for Success

- **Use the 3D cursor** to place objects precisely on the floor or walls.
- **Duplicate and modify** – a book modeled once can be copied, scaled, and rotated to fill a whole shelf.
- **Keep it low-poly but detailed.** You don't need image textures; solid colors and good shapes look amazing in isometric style.
- **Check your camera constantly (The camera icon on the right).** An isometric scene only works from that one angle – if it doesn't look good there, it doesn't exist!